

Qualification Pack

Multimics Data Analyst

QP Code: NIE/SSC/Q1801

Version: 1.0

NSQF Level: 5.5

National Institute of Electronics and Information Technology (NIELIT) || NIELIT Bhawan, Plot No. 3, PSP
Pocket, Sector-8, Dwarka
New Delhi-110077 || email:puneetnema@nielit.gov.in

Qualification Pack

Contents

NIE/SSC/Q1801: Multiomics Data Analyst	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
NIE/SSC/N1805: Use of R Programming in Biological Data Analysis	5
NIE/SSC/N1806: Application of Omics in Big Data Analytics	9
NIE/SSC/N1807: Mathematical Modeling of Biological Networks using Systems Biology Framework ...	13
NIE/SSC/N1808: Healthcare Informatics for Medical Data Management	17
NIE/SSC/N1804: Data security law	21
NIE/SSC/N1809: Implementation of Rprogramming, Omics, data analysis in healthcare informatics- Project	27
DGT/VSQ/N0103: Employability Skills (90 Hours)	31
Assessment Guidelines and Weightage	39
<i>Assessment Guidelines</i>	39
<i>Assessment Weightage</i>	40
Acronyms	42
Glossary	45

Qualification Pack

NIE/SSC/Q1801: Multiomics Data Analyst

Brief Job Description

This Short-term Certificate Course is targeted for creating qualified professionals in the field of Multiomics Data, which will help in employment or Entrepreneurship development of the qualifier in the areas of Health Informatics/ Medical Informatics/ Life Sciences, etc. The individual performs research and designing/reshaping of algorithmic models for identification, troubleshooting, and implementation of improvements to existing pipelines. The person also provides lead scientific, technical and knowledge support for revealing hidden patterns and information in biological big data. S/he is a key contributor in building the drug discovery platform for the future.

Personal Attributes

The purpose of this qualification is to train the students of Life Sciences with advanced-level Bioinformatics and Multiomics Data to upskill them and increase their employability in the field of Life Sciences and allied disciplines.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [NIE/SSC/N1805: Use of R Programming in Biological Data Analysis](#)
2. [NIE/SSC/N1806: Application of Omics in Big Data Analytics](#)
3. [NIE/SSC/N1807: Mathematical Modeling of Biological Networks using Systems Biology Framework](#)
4. [NIE/SSC/N1808: Healthcare Informatics for Medical Data Management](#)
5. [NIE/SSC/N1804: Data security law](#)
6. [NIE/SSC/N1809: Implementation of Rprogramming, Omics, data analysis in healthcare informatics-Project](#)
7. [DGT/VSQ/N0103: Employability Skills \(90 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	IT Services
Occupation	Bioinformatics Data Analytics

Qualification Pack

Country	India
NSQF Level	5.5
Credits	22
Aligned to NCO/ISCO/ISIC Code	5.5
Minimum Educational Qualification & Experience	Completed 3 year UG degree (Completed 3 Years of UG in Biology (Zoology, Botany, Biological Science or related branches)) OR Pursuing 3rd year of UG (in Biology (Zoology, Botany, Biological Science or related branches)*Subject to being offered as 6 Months internship/project, Basic knowledge of computer handling and having knowledge of data crunching.) OR Previous relevant Qualification of NSQF Level (Level5) with 1-2 Years of experience 1.5 year relevant experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	16 Years
Last Reviewed On	NA
Next Review Date	15/03/2027
NSQC Approval Date	15/03/2024
Version	1.0
Reference code on NQR	QG-5.5-IT-02181-2024-V1-NIELIT
NQR Version	1

Qualification Pack

NIE/SSC/N1805: Use of R Programming in Biological Data Analysis

Description

Module introduces R Programming's application in biological data analysis, emphasizing comprehension, link establishment, and practical usage. Students learn R's fundamentals, its role in deciphering biological phenomena, and how to analyze biological data through interactive graphical plots, fostering proficiency in data interpretation. By mastering R, students gain a powerful toolset for extracting insights and visualizing trends, enhancing their capabilities in biological research.

Scope

The scope covers the following :

- Module covers the foundational understanding and practical application of R Programming in biological data analysis. Students will learn R's fundamentals and its relevance in biological research, gaining skills to analyze datasets and create interactive graphical plots. The module aims to equip students with proficiency in using R for insightful data interpretation and unraveling complex biological phenomena.

Elements and Performance Criteria

Introduction to R in Biological Data Analysis & • Bioinformatics Applications in R

To be competent, the user/individual on the job must be able to:

PC1. • Overview of R programming language, Installing R, Choosing and IDE; Application of R in biological data.

•

PC2. Sequence analysis using Bioconductor packages; Genomic data analysis and visualization

Advanced Topics in Bioinformatics with R & • Emerging Trends and Technologies

To be competent, the user/individual on the job must be able to:

PC3. Functional enrichment analysis; Pathway analysis and systems biology

PC4. Application of R in single-cell RNA sequencing; Use of R for analyzing multi-omics data.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. Mastery of R Programming fundamentals, encompassing data structures, syntax, functions, and libraries.

KU2. Recognition of the intrinsic link between R programming and biological data analysis, understanding how R facilitates the manipulation and interpretation of biological datasets.

KU3. Proficiency in utilizing R for the analysis and interpretation of biological data, with a focus on generating interactive graphical plots to visualize trends and patterns effectively.

Generic Skills (GS)

Qualification Pack

User/individual on the job needs to know how to:

- GS1.** Comprehensive understanding of R Programming, including data structures, syntax, functions, and libraries.
- GS2.** Ability to establish and articulate the connection between R programming and biological data analysis, demonstrating insight into how R facilitates data manipulation and interpretation in a biological context.
- GS3.** Proficiency in applying R for the analysis and interpretation of biological data, with a focus on creating interactive graphical plots to effectively communicate findings and insights.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to R in Biological Data Analysis & • Bioinformatics Applications in R</i>	-	-	-	-
PC1. • Overview of R programming language, Installing R, Choosing and IDE; Application of R in biological data. •	-	-	-	-
PC2. Sequence analysis using Bioconductor packages; Genomic data analysis and visualization	-	-	-	-
<i>Advanced Topics in Bioinformatics with R & • Emerging Trends and Technologies</i>	-	-	-	-
PC3. Functional enrichment analysis; Pathway analysis and systems biology	-	-	-	-
PC4. Application of R in single-cell RNA sequencing; Use of R for analyzing multi-omics data.	-	-	-	-
NOS Total	56	25	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/SSC/N1805
NOS Name	Use of R Programming in Biological Data Analysis
Sector	IT-ITeS
Sub-Sector	
Occupation	Bioinformatics Data Analytics
NSQF Level	5.5
Credits	5
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	15/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

NIE/SSC/N1806: Application of Omics in Big Data Analytics

Description

Application of Omics in Big Data Analytics equips students with essential skills in computer programming and Linux, crucial for navigating the complexities of big data analysis. Through this module, students gain a comprehensive understanding of Python programming, enabling them to effectively manipulate and analyze large datasets. By mastering the fundamentals of computer programming and Linux basics, students develop the proficiency needed to tackle real-world challenges in omics and big data analytics. This module empowers learners to apply Python programming techniques confidently, facilitating their journey towards becoming adept data analysts in the field of omics.

Scope

The scope covers the following :

- This NOS encompasses providing students with comprehensive knowledge and practical skills in computer programming fundamentals, Linux basics, and Python application. Through this module, students delve into the intricacies of big data analysis, learning to manipulate and analyze large datasets with Python. By mastering these essential skills, students are prepared to tackle real-world challenges in omics and big data analytics confidently. The scope extends to empowering students to apply Python programming techniques effectively, enabling them to excel as proficient data analysts in the field of omics and contribute meaningfully to advancements in big data analytics.

Elements and Performance Criteria

Introduction to Omics Data in Big Data Analytics & Network Analysis in Omics Big Data

To be competent, the user/individual on the job must be able to:

- PC1.** define the types of omics data (genomics, transcriptomics, proteomics, metabolomics) and Challenges and opportunities in handling large-scale omics datasets
- PC2.** construct and able to do analysis of biological networks, Identification of key nodes and pathways from omics data

Big Data Analytics Platforms for Omics & Precision Medicine

To be competent, the user/individual on the job must be able to:

- PC3.** identify differentially expressed genes, proteins, or metabolites; Machine learning algorithms for omics data classification and prediction

• Precision Medicine and Personalized Omics & • Emerging Technologies and Trends in Omics Big Data Analytics

To be competent, the user/individual on the job must be able to:

- PC4.** • explore large-scale omics datasets.
• They will get Interactive visualization for gaining insights into complex biological systems.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

Qualification Pack

- KU1.** This entails comprehending core concepts like variables, data types, control structures (loops, conditionals), functions, and algorithms. It's about learning how to break down problems into logical steps that a computer can execute, regardless of the programming language used.
- KU2.** Understanding Linux involves familiarity with its command-line interface, file system structure, navigation commands (cd, ls), file manipulation (mkdir, cp, rm), permissions (chmod, chown), and package management (apt, yum).

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.**
- General Skills: Problem-solving, logical thinking, attention to detail.
 - Professional Skills: Proficiency in a programming language, algorithmic thinking, debugging.
- GS2.**
- General Skills: Command-line proficiency, file system navigation.
 - Professional Skills: Shell scripting, system administration tasks.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Omics Data in Big Data Analytics & Network Analysis in Omics Big Data</i>	-	-	-	-
PC1. define the types of omics data (genomics, transcriptomics, proteomics, metabolomics) and Challenges and opportunities in handling large-scale omics datasets	-	-	-	-
PC2. construct and able to do analysis of biological networks, Identification of key nodes and pathways from omics data	-	-	-	-
<i>Big Data Analytics Platforms for Omics & Precision Medicine</i>	-	-	-	-
PC3. identify differentially expressed genes, proteins, or metabolites; Machine learning algorithms for omics data classification and prediction	-	-	-	-
<i>• Precision Medicine and Personalized Omics & • Emerging Technologies and Trends in Omics Big Data Analytics</i>	-	-	-	-
PC4. • explore large-scale omics datasets. • They will get Interactive visualization for gaining insights into complex biological systems.	-	-	-	-
NOS Total	44	20	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/SSC/N1806
NOS Name	Application of Omics in Big Data Analytics
Sector	IT-ITeS
Sub-Sector	
Occupation	Bioinformatics Data Analytics
NSQF Level	5.5
Credits	4
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	15/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

NIE/SSC/N1807: Mathematical Modeling of Biological Networks using Systems Biology Framework

Description

The NOS encompasses a comprehensive study of Systems Biology Framework focusing on mathematical modeling of Biological Networks. It includes an introduction to Systems Biology, various types of Biological Networks, and Mathematical Representation of Biological Components using differential equations and logical modeling.

Scope

The scope covers the following :

- Introduction to Systems Biology and Network Modeling: Fundamentals and applications in biological network analysis, emphasizing mathematical modeling.
- Types of Biological Networks: Study of gene regulatory, protein-protein interaction, metabolic, and signaling networks for structure-function understanding.
- Mathematical Representation and Analysis: Using differential equations, Boolean modeling, and kinetic approaches for network dynamics and topology analysis.
- Applications and Tools: Applying models for drug targets, disease mechanisms, and synthetic biology, with hands-on experience using software like COPASI and CellDesigner.

Elements and Performance Criteria

Fundamentals of Systems Biology and Network Modeling

To be competent, the user/individual on the job must be able to:

- PC1.** Ability to explain the key principles of systems biology and their relevance in biological research.
- PC2.** Demonstrated knowledge of mathematical modeling techniques used in systems biology, such as differential equations and Boolean modeling.
- PC3.** Capability to identify and describe different types of biological networks and their functions in cellular processes.

Analysis of Biological Networks:

To be competent, the user/individual on the job must be able to:

- PC4.** Proficiency in constructing mathematical models for gene regulatory, protein-protein interaction, and metabolic networks.
- PC5.** Competence in applying network analysis methods to study network topology, dynamics, and stability
- PC6.** Skill in integrating experimental data into network models and evaluating model accuracy through validation techniques

Application and Tools

To be competent, the user/individual on the job must be able to:

- PC7.** Ability to apply mathematical models to simulate dynamic behaviors and predict system responses.

Qualification Pack

- PC8.** Competency in using software tools like COPASI and CellDesigner for modeling biological networks
- PC9.** Capability to apply systems biology models in practical scenarios such as drug target identification and disease mechanism understanding.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.**
- Knowledge of Systems Biology Principles:
 - Understanding of systems biology concepts and their applications in studying biological systems.
 - Familiarity with mathematical modeling techniques used to represent biological networks.
 - Knowledge of different types of biological networks and their roles in cellular processes.
- KU2.**
- Understanding of Network Modeling:
 - Understanding of how to construct mathematical models for various biological networks.
 - Knowledge of network analysis methods for studying network topology, dynamics, and stability.
 - Understanding of data integration techniques to refine and validate network models.
- KU3.**
- Application of Systems Biology Models:
 - Knowledge of applying mathematical models to simulate dynamic behaviors of biological systems.
 - Understanding of using systems biology models in practical applications such as drug target identification and disease mechanism investigation.
 - Familiarity with software tools used for systems biology modeling and analysis, such as COPASI and CellDesigner

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.**
- Mathematical Modeling Skills:
 - Ability to create mathematical models to represent complex biological networks.
 - Proficiency in using differential equations, Boolean logic, and kinetic modeling for network representation.
 - Skill in parameter estimation and model validation techniques to ensure model accuracy.
- GS2.**
- Data Integration and Analysis Skills:
 - Capability to integrate experimental data, including omics data, into network models.
 - Proficiency in data assimilation techniques to refine and update mathematical models.
 - Skill in analyzing network topology and dynamic behavior using computational tools.
- GS3.**
- Software Proficiency:
 - Competence in using software tools specific to systems biology modeling, such as COPASI and CellDesigner.
 - Ability to perform network simulations, stability analysis, and data integration within modeling platforms.
 - Skill in comparative analysis of software tools to choose the most suitable for different modeling tasks.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Fundamentals of Systems Biology and Network Modeling</i>	-	-	-	-
PC1. Ability to explain the key principles of systems biology and their relevance in biological research.	-	-	-	-
PC2. Demonstrated knowledge of mathematical modeling techniques used in systems biology, such as differential equations and Boolean modeling.	-	-	-	-
PC3. Capability to identify and describe different types of biological networks and their functions in cellular processes.	-	-	-	-
<i>Analysis of Biological Networks:</i>	-	-	-	-
PC4. Proficiency in constructing mathematical models for gene regulatory, protein-protein interaction, and metabolic networks.	-	-	-	-
PC5. Competence in applying network analysis methods to study network topology, dynamics, and stability	-	-	-	-
PC6. Skill in integrating experimental data into network models and evaluating model accuracy through validation techniques	-	-	-	-
<i>Application and Tools</i>	-	-	-	-
PC7. Ability to apply mathematical models to simulate dynamic behaviors and predict system responses.	-	-	-	-
PC8. Competency in using software tools like COPASI and CellDesigner for modeling biological networks	-	-	-	-
PC9. Capability to apply systems biology models in practical scenarios such as drug target identification and disease mechanism understanding.	-	-	-	-
NOS Total	50	23	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/SSC/N1807
NOS Name	Mathematical Modeling of Biological Networks using Systems Biology Framework
Sector	IT-ITeS
Sub-Sector	
Occupation	Bioinformatics Data Analytics
NSQF Level	5.5
Credits	4
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	15/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

NIE/SSC/N1808: Healthcare Informatics for Medical Data Management

Description

The NOS encompasses a wide array of topics crucial to modern healthcare systems. It starts with an introduction to the field, highlighting its scope and importance in managing medical data efficiently. The focus then shifts to Electronic Health Records (EHR) Systems, Health Information Exchange (HIE), and Clinical Decision Support Systems (CDSS), emphasizing seamless data exchange, privacy, security, and decision-making support.

Scope

The scope covers the following :

- Fundamentals of Healthcare Informatics: Understanding the scope and significance of healthcare informatics in managing medical data efficiently.
- Electronic Health Records (EHR) Systems: Exploring the design, implementation, and interoperability standards of electronic health record systems for seamless data exchange.
- Health Information Exchange (HIE): Learning about connecting disparate healthcare systems, focusing on information sharing protocols and addressing privacy and security concerns.
- Clinical Decision Support Systems (CDSS): Integrating informatics into clinical decision-making processes, emphasizing the applications and advantages of CDSS in healthcare settings.

Elements and Performance Criteria

Understanding Healthcare Informatics Fundamentals

To be competent, the user/individual on the job must be able to:

- PC1.** Demonstrate an understanding of healthcare informatics by defining key terms and explaining their relevance in healthcare.
- PC2.** Identify and discuss the challenges and benefits associated with effective medical data management in healthcare settings
- PC3.** Apply knowledge of healthcare informatics fundamentals to analyze real-world scenarios and propose solutions for improving data management processes.

Exploring Electronic Health Records (EHR) Systems

To be competent, the user/individual on the job must be able to:

- PC4.** Design and develop a basic electronic health record system prototype, including data storage and retrieval functionalities
- PC5.** Evaluate interoperability standards and protocols to ensure seamless data exchange between different EHR systems.
- PC6.** Assess the impact of EHR systems on patient care and healthcare workflows through case studies and simulations.

Health Information Exchange (HIE) and Clinical Decision Support Systems (CDSS)

To be competent, the user/individual on the job must be able to:

- PC7.** Implement a secure health information exchange mechanism using encryption and access control measures.

Qualification Pack

- PC8.** Evaluate the effectiveness of CDSS in aiding clinical decision-making by analyzing data-driven recommendations and their outcomes.
- PC9.** Develop and present a comprehensive strategy for ensuring privacy and security in health information exchange processes, considering regulatory requirements and best practices.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** • Healthcare Informatics Fundamentals:
• Understand the core concepts and principles of healthcare informatics, including the importance of medical data management, interoperability standards, and data security.
- KU2.** • Electronic Health Records (EHR) Systems:
• Gain knowledge about the design, implementation, and functionalities of electronic health record systems, including data storage, retrieval, and exchange.
- KU3.** • Health Information Exchange (HIE) and Clinical Decision Support Systems (CDSS):
• Learn about health information exchange mechanisms, privacy considerations, and the role of clinical decision support systems in enhancing healthcare decision-making processes.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Data Management: Develop skills in managing and organizing medical data efficiently, including electronic health records, data exchange protocols, and data security practices.
- GS2.** Problem-Solving: Enhance problem-solving abilities by applying healthcare informatics principles to address challenges such as interoperability issues, data standardization, and privacy concerns in health information exchange
- GS3.** Communication: Improve communication skills by effectively conveying complex healthcare informatics concepts, data management strategies, and regulatory compliance requirements to stakeholders such as healthcare providers, IT professionals, and policymakers.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Understanding Healthcare Informatics Fundamentals</i>	-	-	-	-
PC1. Demonstrate an understanding of healthcare informatics by defining key terms and explaining their relevance in healthcare.	-	-	-	-
PC2. Identify and discuss the challenges and benefits associated with effective medical data management in healthcare settings	-	-	-	-
PC3. Apply knowledge of healthcare informatics fundamentals to analyze real-world scenarios and propose solutions for improving data management processes.	-	-	-	-
<i>Exploring Electronic Health Records (EHR) Systems</i>	-	-	-	-
PC4. Design and develop a basic electronic health record system prototype, including data storage and retrieval functionalities	-	-	-	-
PC5. Evaluate interoperability standards and protocols to ensure seamless data exchange between different EHR systems.	-	-	-	-
PC6. Assess the impact of EHR systems on patient care and healthcare workflows through case studies and simulations.	-	-	-	-
<i>Health Information Exchange (HIE) and Clinical Decision Support Systems (CDSS)</i>	-	-	-	-
PC7. Implement a secure health information exchange mechanism using encryption and access control measures.	-	-	-	-
PC8. Evaluate the effectiveness of CDSS in aiding clinical decision-making by analyzing data-driven recommendations and their outcomes.	-	-	-	-
PC9. Develop and present a comprehensive strategy for ensuring privacy and security in health information exchange processes, considering regulatory requirements and best practices.	-	-	-	-
NOS Total	25	11	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/SSC/N1808
NOS Name	Healthcare Informatics for Medical Data Management
Sector	IT-ITeS
Sub-Sector	
Occupation	Bioinformatics Data Analytics
NSQF Level	5.5
Credits	2
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	15/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

NIE/SSC/N1804: Data security law

Description

The course on Data Security Law covers a comprehensive range of topics essential for understanding and implementing effective data protection measures. Students will gain insights into the evolution of data security laws, including key concepts and terminology.

Scope

The scope covers the following :

- **Foundations of Data Security Law:** Covers fundamental concepts, historical evolution, and the importance of data protection.
- **Legal Frameworks and Compliance:** Focuses on major data protection laws, comparative analysis, and compliance challenges.
- **Managing Data Breaches:** Addresses understanding breaches, legal responses, and developing incident plans.
- **Risk Management and Emerging Trends:** Includes risk assessment, compliance strategies, and future developments in data security laws.

Elements and Performance Criteria

Conceptual Understanding

To be competent, the user/individual on the job must be able to:

- PC1.** Demonstrate comprehension of key data security concepts and terminology by accurately defining and explaining terms like encryption, authentication, and data breach.
- PC2.** Identify and describe the historical evolution of data security laws, including landmark legislation and significant events that shaped data protection practices.
- PC3.** Evaluate the importance of data protection in modern society by analyzing case studies, industry reports, and real-world examples to understand the impact of data breaches on businesses and individuals.

Legal Framework Exploration

To be competent, the user/individual on the job must be able to:

- PC4.** Compare and contrast major data protection laws such as GDPR, CCPA, HIPAA, and analyze their scope, requirements, and implications for businesses and consumers.
- PC5.** Critically examine international data security regulations to understand the global landscape of data protection and privacy laws, including differences in approaches and enforcement mechanisms.
- PC6.** Identify and explain jurisdictional differences and compliance challenges related to data security laws, considering factors like cross-border data transfers, data residency requirements, and regulatory oversight.

Practical Application and Response

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** Develop and propose effective strategies for data controllers and processors to comply with data security laws, including data subject rights management, consent mechanisms, and data protection impact assessments.
- PC8.** Create incident response plans and protocols for managing data breaches and security incidents, including steps for detection, containment, mitigation, notification, and recovery.
- PC9.** Evaluate and recommend risk management and compliance strategies, including risk assessment methodologies, technical and organizational measures, and best practices for ensuring data security and regulatory compliance.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.**
- Comprehensive Understanding of Data Security Concepts:
 - Knowledge of fundamental data security concepts and terminology, including encryption methods, authentication mechanisms, data breach prevention strategies, and cybersecurity best practices.
 - Understanding the historical evolution of data security laws, including the development of key legislation such as GDPR, HIPAA, and CCPA, and their impact on data protection practices globally.
 - Awareness of the importance of data protection in modern soc
- KU2.**
- Legal Framework and Compliance Knowledge:
 - Familiarity with major data protection laws and regulations, such as GDPR, CCPA, HIPAA, and their respective legal frameworks, requirements, and enforcement mechanisms.
 - Understanding of international data security regulations and jurisdictional differences, including comparative analysis of data protection laws across different regions and countries.
 - Knowledge of compliance requirements and legal obligations for data controllers and processors, including
- KU3.**
- Practical Application and Response Strategies:
 - Application of legal principles and requirements to real-world scenarios, such as developing compliance strategies, drafting privacy policies, and implementing data protection measures.
 - Understanding of incident response management, including identifying data breaches, assessing the impact, notifying affected parties, and coordinating with regulatory authorities.
 - Knowledge of risk management strategies, including risk assessment methodologies, data

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.**
- Legal Research and Analysis:
 - Ability to conduct in-depth legal research on data security laws, regulations, and compliance requirements.
 - Analytical skills to interpret and analyze complex legal frameworks, including understanding jurisdictional differences and compliance challenges.
 - Capability to synthesize legal information and apply it to practical scenarios, such as assessing the legal implications of data breaches and developing compliance strategies.

Qualification Pack

- GS2.** • Communication and Documentation:
- Effective communication skills to convey complex legal concepts, compliance requirements, and risk assessments to various stakeholders within an organization.
 - Proficiency in drafting clear and concise legal documents, including privacy policies, data protection agreements, and incident response plans.
 - Capability to communicate with regulatory authorities, legal teams, and data protection officers regarding compliance issues, data breach notifications, and regulat
- GS3.** • Problem-Solving and Decision-Making:
- Critical thinking skills to analyze data security challenges, identify potential risks and vulnerabilities, and develop proactive risk management strategies.
 - Ability to make informed decisions regarding data protection measures, incident response protocols, and compliance initiatives based on legal requirements and risk assessments.
 - Problem-solving skills to address data breaches, security incidents, and regulatory compliance issues effectively, including dev

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Conceptual Understanding</i>	-	-	-	-
PC1. Demonstrate comprehension of key data security concepts and terminology by accurately defining and explaining terms like encryption, authentication, and data breach.	-	-	-	-
PC2. Identify and describe the historical evolution of data security laws, including landmark legislation and significant events that shaped data protection practices.	-	-	-	-
PC3. Evaluate the importance of data protection in modern society by analyzing case studies, industry reports, and real-world examples to understand the impact of data breaches on businesses and individuals.	-	-	-	-
<i>Legal Framework Exploration</i>	-	-	-	-
PC4. Compare and contrast major data protection laws such as GDPR, CCPA, HIPAA, and analyze their scope, requirements, and implications for businesses and consumers.	-	-	-	-
PC5. Critically examine international data security regulations to understand the global landscape of data protection and privacy laws, including differences in approaches and enforcement mechanisms.	-	-	-	-
PC6. Identify and explain jurisdictional differences and compliance challenges related to data security laws, considering factors like cross-border data transfers, data residency requirements, and regulatory oversight.	-	-	-	-
<i>Practical Application and Response</i>	-	-	-	-
PC7. Develop and propose effective strategies for data controllers and processors to comply with data security laws, including data subject rights management, consent mechanisms, and data protection impact assessments.	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC8. Create incident response plans and protocols for managing data breaches and security incidents, including steps for detection, containment, mitigation, notification, and recovery.	-	-	-	-
PC9. Evaluate and recommend risk management and compliance strategies, including risk assessment methodologies, technical and organizational measures, and best practices for ensuring data security and regulatory compliance.	-	-	-	-
NOS Total	25	11	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/SSC/N1804
NOS Name	Data security law
Sector	IT-ITeS
Sub-Sector	
Occupation	Bioinformatics Data Analytics
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	15/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

NIE/SSC/N1809: Implementation of Rprogramming, Omics, data analysis in healthcare informatics- Project

Description

This National Occupational Standard (NOS) defines the essential skills for professionals in multiomics data analysis, covering R programming, big data analytics, and biological network modeling. It includes data preprocessing, statistical modeling, machine learning, and healthcare informatics. The curriculum also addresses systems biology, data management, regulatory compliance, and cybersecurity policies. Learners gain expertise in analyzing complex biological datasets, building predictive models, and using advanced bioinformatics tools. This comprehensive approach ensures secure and ethical data handling, preparing professionals to tackle real-world challenges in biomedical and multiomics data science.

Scope

The scope covers the following :

- The scope of the National Occupational Standard (NOS) for Multiomics Data Analysis includes expertise in R programming, big data analytics, and biological network modeling. It covers data preprocessing, statistical modeling, machine learning, and healthcare informatics. Emphasis is placed on regulatory compliance, cybersecurity policies, and ethical data management. Professionals are prepared to analyze complex biological datasets, develop predictive models, and utilize advanced bioinformatics tools.

Elements and Performance Criteria

Project: Multiomics Data Analysis and Integration

To be competent, the user/individual on the job must be able to:

- PC1.** Utilize R programming for preprocessing, analyzing, and interpreting biological datasets in multiomics research
- PC2.** Implement big data analytics and mathematical modeling techniques to study complex biological networks.
- PC3.** Develop machine learning-based predictive models and integrate systems biology frameworks for healthcare applications.
- PC4.** Ensure compliance with regulatory standards, cybersecurity policies, and ethical guidelines for secure data handling.
- PC5.** Apply advanced bioinformatics tools for data integration, visualization, and problem-solving in biomedical research.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Fundamental concepts of R programming, data preprocessing, and exploratory data analysis for multiomics research

Qualification Pack

- KU2.** Techniques for applying big data analytics and mathematical modeling to study biological networks.
- KU3.** Machine learning applications in predictive modeling and integration of systems biology frameworks.
- KU4.** Regulatory compliance, cybersecurity policies, and ethical considerations for secure data handling.
- KU5.** Advanced bioinformatics tools for data integration, visualization, and biomedical problem-solving.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Develop proficiency in R programming, data preprocessing, and exploratory data analysis for multiomics research. Acquire the ability to clean, transform, and visualize complex biological datasets efficiently.
- GS2.** Strengthen analytical skills to apply big data techniques and mathematical modeling for biological network analysis. Utilize machine learning to develop predictive models and interpret results for biomedical applications.
- GS3.** Effectively communicate technical concepts related to multiomics data analysis. Prepare clear documentation on methodologies, data insights, and regulatory compliance to ensure seamless collaboration with researchers and stakeholders.
- GS4.** Apply knowledge of data security laws, cybersecurity policies, and ethical considerations to ensure secure and compliant handling of biological and healthcare data.

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Project: Multiomics Data Analysis and Integration</i>	-	-	-	-
PC1. Utilize R programming for preprocessing, analyzing, and interpreting biological datasets in multiomics research	-	-	-	-
PC2. Implement big data analytics and mathematical modeling techniques to study complex biological networks.	-	-	-	-
PC3. Develop machine learning-based predictive models and integrate systems biology frameworks for healthcare applications.	-	-	-	-
PC4. Ensure compliance with regulatory standards, cybersecurity policies, and ethical guidelines for secure data handling.	-	-	-	-
PC5. Apply advanced bioinformatics tools for data integration, visualization, and problem-solving in biomedical research.	-	-	-	-
NOS Total	-	-	160	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/SSC/N1809
NOS Name	Implementation of Rprogramming, Omics, data analysis in healthcare informatics- Project
Sector	IT-ITeS
Sub-Sector	
Occupation	Bioinformatics Data Analytics
NSQF Level	5.5
Credits	2
Version	1.0
Last Reviewed Date	15/03/2024
Next Review Date	15/03/2027
NSQC Clearance Date	15/03/2024

Qualification Pack

DGT/VSQ/N0103: Employability Skills (90 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** understand the significance of employability skills in meeting the current job market requirement and future of work
- PC2.** identify and explore learning and employability relevant portals
- PC3.** research about the different industries, job market trends, latest skills required and the available opportunities

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC4.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC5.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC6.** recognize the significance of 21st Century Skills for employment

Qualification Pack

- PC7.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life
- PC8.** adopt a continuous learning mindset for personal and professional development

Basic English Skills

To be competent, the user/individual on the job must be able to:

- PC9.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC10.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC11.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC12.** identify career goals based on the skills, interests, knowledge, and personal attributes
- PC13.** prepare a career development plan with short- and long-term goals

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC14.** follow verbal and non-verbal communication etiquette while communicating in professional and public settings
- PC15.** use active listening techniques for effective communication
- PC16.** communicate in writing using appropriate style and format based on formal or informal requirements
- PC17.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC18.** communicate and behave appropriately with all genders and PwD
- PC19.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC20.** identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.
- PC21.** carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook
- PC22.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC23.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC24.** operate digital devices and use their features and applications securely and safely
- PC25.** carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.
- PC26.** display responsible online behaviour while using various social media platforms

Qualification Pack

- PC27.** create a personal email account, send and process received messages as per requirement
- PC28.** carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications
- PC29.** utilize virtual collaboration tools to work effectively

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC30.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC31.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC32.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC33.** identify different types of customers and ways to communicate with them
- PC34.** identify and respond to customer requests and needs in a professional manner
- PC35.** use appropriate tools to collect customer feedback
- PC36.** follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

- PC37.** create a professional Curriculum vitae (Résumé)
- PC38.** search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively
- PC39.** apply to identified job openings using offline /online methods as per requirement
- PC40.** answer questions politely, with clarity and confidence, during recruitment and selection
- PC41.** identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** need for employability skills and different learning and employability related portals
- KU2.** various constitutional and personal values
- KU3.** different environmentally sustainable practices and their importance
- KU4.** Twenty first (21st) century skills and their importance
- KU5.** how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up
- KU6.** importance of career development and setting long- and short-term goals
- KU7.** about effective communication
- KU8.** POSH Act
- KU9.** Gender sensitivity and inclusivity
- KU10.** different types of financial institutes, products, and services

Qualification Pack

- KU11.** components of salary and how to compute income and expenditure
- KU12.** importance of maintaining safety and security in offline and online financial transactions
- KU13.** different legal rights and laws
- KU14.** different types of digital devices and the procedure to operate them safely and securely
- KU15.** how to create and operate an e- mail account
- KU16.** use applications such as word processors, spreadsheets etc.
- KU17.** how to identify business opportunities
- KU18.** types and needs of customers
- KU19.** how to apply for a job and prepare for an interview
- KU20.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and write different types of documents/instructions/correspondence in English and other languages
- GS2.** communicate effectively using appropriate language in formal and informal settings
- GS3.** behave politely and appropriately with all to maintain effective work relationship
- GS4.** how to work in a virtual mode, using various technological platforms
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the current job market requirement and future of work	-	-	-	-
PC2. identify and explore learning and employability relevant portals	-	-	-	-
PC3. research about the different industries, job market trends, latest skills required and the available opportunities	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC4. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC5. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC6. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC7. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
PC8. adopt a continuous learning mindset for personal and professional development	-	-	-	-
<i>Basic English Skills</i>	3	4	-	-
PC9. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC11. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-
PC12. identify career goals based on the skills, interests, knowledge, and personal attributes	-	-	-	-
PC13. prepare a career development plan with short- and long-term goals	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC14. follow verbal and non-verbal communication etiquette while communicating in professional and public settings	-	-	-	-
PC15. use active listening techniques for effective communication	-	-	-	-
PC16. communicate in writing using appropriate style and format based on formal or informal requirements	-	-	-	-
PC17. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC18. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC19. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC20. identify and select reliable institutions for various financial products and services such as bank account, debit and credit cards, loans, insurance etc.	-	-	-	-
PC21. carry out offline and online financial transactions, safely and securely, using various methods and check the entries in the passbook	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC22. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC23. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	5	-	-
PC24. operate digital devices and use their features and applications securely and safely	-	-	-	-
PC25. carry out basic internet operations by connecting to the internet safely and securely, using the mobile data or other available networks through Bluetooth, Wi-Fi, etc.	-	-	-	-
PC26. display responsible online behaviour while using various social media platforms	-	-	-	-
PC27. create a personal email account, send and process received messages as per requirement	-	-	-	-
PC28. carry out basic procedures in documents, spreadsheets and presentations using respective and appropriate applications	-	-	-	-
PC29. utilize virtual collaboration tools to work effectively	-	-	-	-
<i>Entrepreneurship</i>	2	3	-	-
PC30. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC31. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC32. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC33. identify different types of customers and ways to communicate with them	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC34. identify and respond to customer requests and needs in a professional manner	-	-	-	-
PC35. use appropriate tools to collect customer feedback	-	-	-	-
PC36. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC37. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC38. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC39. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC40. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC41. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0103
NOS Name	Employability Skills (90 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	5
Credits	3
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

- Assessment of the qualification evaluates candidates to ascertain that they can integrate knowledge, skills and values for carrying out relevant tasks as per the
- defined learning outcomes and assessment criteria.
- The underlying principle of assessment is fairness and transparency. The evidence of the outcomes and assessment criteria. competence acquired by the candidate • can be obtained by conducting Theory (Online), Practical assessment, Internal assessment, Project/Presentation/ Assignment, Major Project. The emphasis is on the
- practical demonstration of skills & knowledge gained by the candidate through the training. Each OUTCOME is assessed & marked separately. A candidate is
- required to pass all OUTCOMES individually based on the passing criteria.

About Examination Pattern:

1. The question papers for the theory and practical exams are set by the Examination wing (assessor) of NIELIT HQS.

Qualification Pack

2. The assessor assigns roll number
3. The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.
4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject expert including NIELIT officials.
5. Pass percentage would be 50% marks in each component.
6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).
7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.
8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.
- 9. The examination will be conducted in English language only.
- Quality assurance activities: A pool of questions is created by a subject matter expert and moderated by other SME. Test rules are set beforehand. Random set of questions which are according to syllabus appears which may differ from candidate to candidate. Confidentiality and impartiality are maintained during all the examination and evaluation processes.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/SSC/N1805.Use of R Programming in Biological Data Analysis	56	25	0	0	81	16

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/SSC/N1806.Application of Omics in Big Data Analytics	44	20	0	0	64	13
NIE/SSC/N1807.Mathematical Modeling of Biological Networks using Systems Biology Framework	50	23	0	0	73	14
NIE/SSC/N1808.Healthcare Informatics for Medical Data Management	25	11	0	0	36	8
NIE/SSC/N1804.Data security law	25	11	0	0	36	7
NIE/SSC/N1809.Implementation of Rprogramming, Omics, data analysis in healthcare informatics- Project	-	-	160	-	160	32
DGT/VSQ/N0103.Employability Skills (90 Hours)	20	30	-	-	50	10
Total	220	120	160	-	500	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework
NOS	National Occupational Standard(s)
NQR	National Qualification Register
NSQF	National Skills Qualifications Framework
OJT	On the Job Training
QP	Qualification Pack
NSQF	National Skills Qualification Framework
NSQC	National Skills Qualification Committee
NOS	National Occupational Standards
AA	Assessment Agency
AB	Awarding Body
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework
NOS	National Occupational Standard(s)

Qualification Pack

NQR	National Qualification Register
NSQF	National Skills Qualification Framework
OJT	OJT On the Job Training
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework
NOS	National Occupational Standard(s)
NQR	National Qualification Register
NSQF	National Skills Qualifications Framework
OJT	On the Job Training
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework
NOS	National Occupational Standard(s)
NQR	National Qualification Register
NSQF	National Skills Qualifications Framework
OJT	On the Job Training
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework

Qualification Pack

NOS	National Occupational Standard(s)
NQR	National Qualification Register
NSQF	National Skills Qualifications Framework
OJT	On the Job Training
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework
NOS	National Occupational Standard(s)
NQR	National Qualification Register
NSQF	National Skills Qualifications Framework
OJT	On the Job Training
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework
NOS	National Occupational Standard(s)
NQR	National Qualification Register
NSQF	National Skills Qualifications Framework
OJT	On the Job Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

Qualification Pack

Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
National Occupational Standard	NOS define the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Sector	A grouping of professional activities on the basis of their main economic function, product, service or technology.
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above. https://ncvet.gov.in/sites/default/files/NCVET.pdf
National Occupational Standard	NOS define the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards

Qualification Pack

Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Sector	A grouping of professional activities on the basis of their main economic function, product, service or technology.
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above. https://ncvet.gov.in/sites/default/files/NCVET.pdf
National Occupational Standard	NOS define the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Sector	A grouping of professional activities on the basis of their main economic function, product, service or technology.
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above. https://ncvet.gov.in/sites/default/files/NCVET.pdf
National Occupational Standard	NOS define the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Sector	A grouping of professional activities on the basis of their main economic function, product, service or technology
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above. https://ncvet.gov.in/sites/default/files/NCVET.pdf

Qualification Pack

National Occupational Standard	NOS define the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Sector	A grouping of professional activities on the basis of their main economic function, product, service or technology.
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above. https://ncvet.gov.in/sites/default/files/NCVET.pdf
National Occupational Standard	NOS define the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Sector	A grouping of professional activities on the basis of their main economic function, product, service or technology.
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above. https://ncvet.gov.in/sites/default/files/NCVET.pdf
National Occupational Standard	NOS define the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.

Qualification Pack

Sector	A grouping of professional activities on the basis of their main economic function, product, service or technology
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above. https://ncvet.gov.in/sites/default/files/NCVET.pdf
National Occupational Standard	NOS define the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Sector	A grouping of professional activities on the basis of their main economic function, product, service or technology
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above. https://ncvet.gov.in/sites/default/files/NCVET.pdf